SAR-RAG Research Status - Corrected Screening and V2 Recalibration

Critical engineering audit | training-only retrieval evidence | 286-example corrected screening + 225-example disjoint v2 recalibration | **zero LLM/Groq calls** | 5 September 2026

Executive conclusion

Question	Verified answer
Is RRF one-or-two-view routing supported?	**No.** Candidate Oracle@2 adds 0 over Oracle@1; the attainable routing ceiling is too small.
Did adaptive retrieval survive?	**Yes, in scope.** The selected v2 retriever strongly beats the shallow probe.
What did the frozen v2 rule choose?	Paragraph, 1,600-token operating point; relevance percentile 0.45; patience 1.
Does v2 replace v1?	**No.** Frozen v1 point-dominates v2 on mean recall and actual tokens; neither paired difference is significant.
Overall v2 category	**UNSUPPORTED as a replacement claim.** Scoped probe/fixed-k trade-offs remain informative.
Held-out answer quality?	**Unknown and blocked.** No paid or held-out answer-generation run occurred.

[DECISION] The paper-safe result is a negative routing finding plus a scoped adaptive-retrieval finding. Development data do not establish answer-quality superiority, and the v2 single-view method does not outperform frozen v1 at the observed recall-token point.

Evidence labels

Label	Meaning
RAW	Directly recomputed from per-example artifacts
STAT	Paired test under the frozen analysis
INFER	Interpretation beyond a direct measurement
CORRECTION	Repairs an earlier population, wording, or comparison error
BLOCKED	Not run; requires explicit authorization

1. Original hypothesis and version-1 pipeline

The original hypothesis was that four structured representations - sentence window, paragraph, semantic, and hierarchical - contain complementary evidence, and that normalized reciprocal-rank fusion can select one or two useful views per query before adaptive retrieval.

V1 stage	Operation	LLM use
1	Parse documents into canonical structure	None
2	Generate and embed four representations with all-MiniLM-L6-v2	None
3	Probe top three per representation; normalized RRF chooses one/two views	None
4	Adaptive retrieval with relevance, novelty, gap, utility, redundancy, patience and caps	None
5	Post-retrieval deduplication	None
6	One final Groq answer-generation call	One - not used in these studies

2. Screening design and corrected audit

Check	Verified value
Raw screening table	9,724 rows = 286 questions x 34 conditions; no duplicate keys
Partitions	194 tuning / 92 selection
Evidence-scored	210 total / 143 tuning / 67 selection
Expected undefined metric cells	2,584 per evidence metric = 76 no-gold questions x 34 conditions
Frozen v1 manifest	5c570ec2b4c52e1b0ea0f2152a6f8fb6f1fa1c1575e58cfb58bde2c434a54d05
Working tree	Not clean; only the frozen-manifest hash is asserted unchanged

[CORRECTION] Answerable CER now uses 46 scored selection questions. The 21 MuSiQue insufficient questions with residual support are reported separately. They are never pooled into the primary CER denominator.

[CORRECTION] The former budget Boolean was ambiguous. Absence of a significantly superior traditional arm does not establish a significant SAR advantage. Comparisons now report actual tokens, matched actual-token points, and a recall-token Pareto analysis.

3. Corrected screening result: routing rejected

Scope	n	Oracle@1	Oracle@2	Pair gain	Positive pair-gain examples
Candidate selection	67	0.7575	0.7575	0.000000	0
Candidate pooled	210	0.8147	0.8147	0.000000	0
Legacy pooled	210	0.8190	0.8196	0.000680	1

[CORRECTION] Pair complementarity is exactly zero for the hierarchical-v2 candidate family across all 210 scored examples and for legacy selection. Legacy pooled data contain one tuning example with gain 0.142857; pooled mean gain is 0.000680. "Zero in every scope" was too broad.

Component observation	Verified result
Novelty removal	0/286 final outputs changed
Similarity-gap removal	0/286 final outputs changed; threshold unreachable in screening
Marginal utility / redundancy	0 candidates bound by either gate
Deduplication	Changed k/tokens on 56/286; recall changed on 0 - efficiency only
Failure reachability	18 of 26 full-system complete misses unreachable by tested candidate singles/pairs

[INFER] Most complete misses are unreachable under the tested encoder, chunking, representation family, budget, and evidence matching. This does not prove the encoder is universally the dominant failure or that routing can never work elsewhere.

4. Adaptive Single-View V2: frozen design

Design element	Frozen specification
Data	Official QASPER train and MuSiQue-Full train only
Fresh sample	225 questions, nine evidence-bearing strata, seed 20260905
Partitions	162 tuning / 63 selection, assigned before retrieval outcomes
Prior training IDs excluded	475; overlap = 0
Held-out ID-only registry	7,291 validation/test identifiers; overlap = 0; content/labels not used
Candidate representations	semantic, paragraph, hierarchical_v2, sentence-window
Token points	800, 1,200, 1,600; 2,000 is a safety maximum, not a target
Adaptive controls	depth-3 probe, batch 3, percentile relevance, patience, token/round/candidate/chunk caps
Removed from primary v2	RRF/router/pairs, novelty, similarity gap, marginal utility, redundancy
Selection	Tuning picks one tuple per representation/token point; frozen selection rule chooses among 12

[RAW] The run evaluated 162 x 1,152 = 186,624 tuning configuration-question rows and 63 x 49 = 3,087 selection rows. The selected manifest reproduced identically on all substantive fields. Final k remained emergent. No Groq or other LLM call occurred.

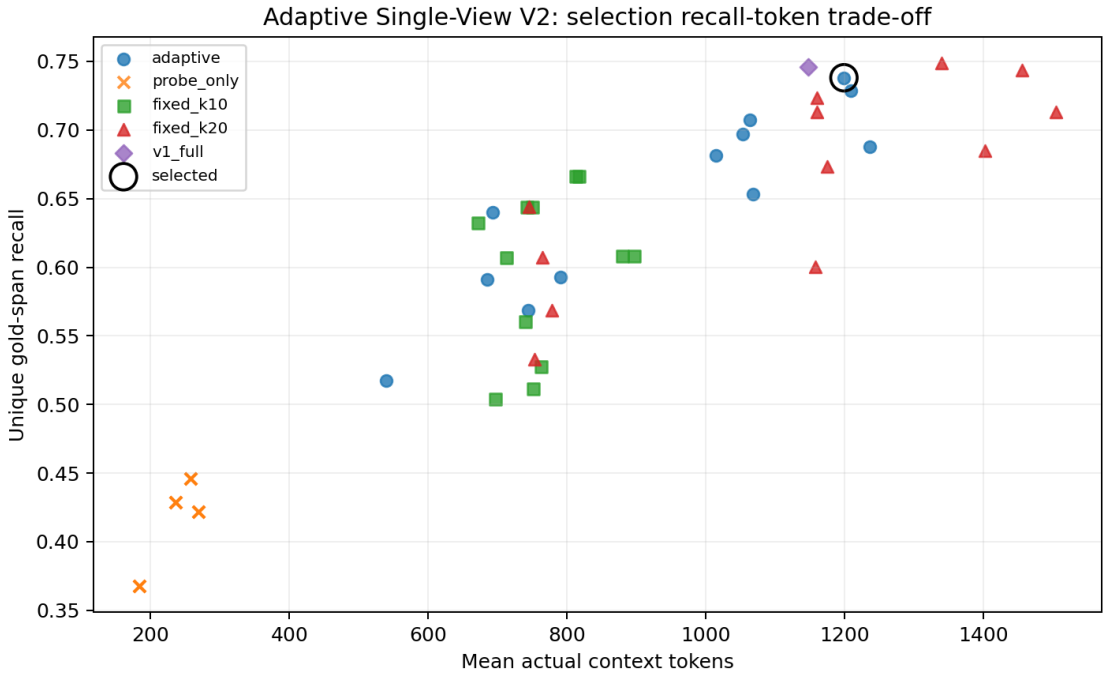
5. Selected v2 operating point

Representation	Token point	Rel. percentile	Patience	Round cap	Candidate cap	Chunk cap
paragraph	1,600	0.45	1	6	30	20

Selection metric	Value	Population / caveat
Unique gold-span recall	0.7381	63 evidence-scored training-selection questions
Answerable CER	0.5000	42 answerable questions only
Residual-support complete recovery	0.8095	21 insufficient questions; separate secondary metric
Mean actual context	1199.2 tokens	75% of 1,600 point
Mean final k	15.00	emergent range 7-20
Candidates / rounds	17.90 / 5.06	means

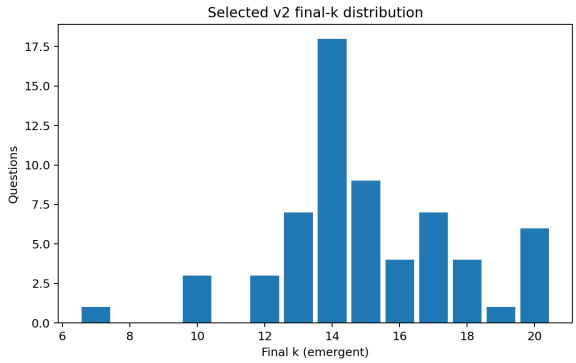
6. Quality-efficiency trade-off

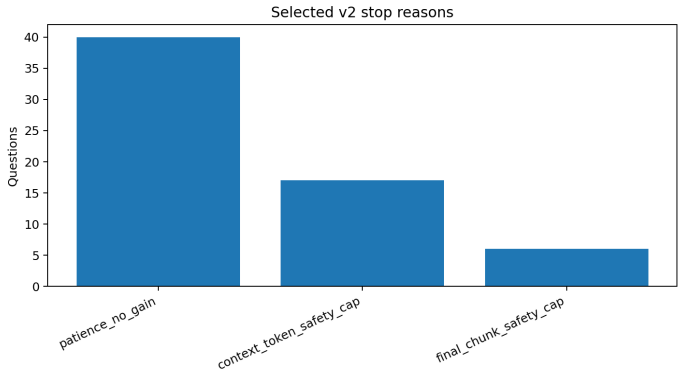
Comparator	Recall difference (v2-ref)	95% CI	Holm p	Token difference	Interpretation
Same-view probe	+0.2923	[+0.2143, +0.3730]	0.00015	+941.2	Large retrieval gain; large context increase
Same-view k=10	+0.0714	[+0.0317, +0.1190]	0.0022	+382.4	Higher recall, but consumes more
Same-view k=20	-0.0053	[-0.0159, +0.0000]	1	-256.8	Recall-noninferior; 17.6% fewer tokens



[DECISION] V2 is non-dominated when the descriptive v1 point is excluded. With v1 included, v1 is the sole point-estimate dominator: recall 0.7460 vs 0.7381 and actual tokens 1,147.6 vs 1,199.2. Paired v2-minus-v1 recall is -0.0079, CI [-0.0675, +0.0476]; token difference is +51.6, CI [-2.9, +100.3]. Neither is significant. Different nominal maxima further limit a formal claim.

7. Behavior, subgroup limitations, and failures





Selection subgroups

Group	n	Recall	Complete recovery	Tokens	Final k
musique_full	42	0.8929	0.7619	1166.4	14.52
qasper	21	0.4286	0.2857	1264.8	15.95
answerable	42	0.6587	0.5000	1246.5	15.36
insufficient	21	0.8968	0.8095	1104.7	14.29

[LIMITATION] QASPER recall is 0.4286 (n=21) versus MuSiQue 0.8929 (n=42). These small, stratified development cells are not standalone evidence. The contrast is a failure-analysis signal, not a dataset-general conclusion.

Failure class	Count / 63
Complete recovery	38
Partial recovery	15
Total miss	10

8. Limitations, blocked paid work, and paper-safe conclusion

Limitation	Consequence
Training selection is design-exposed	All v2 results are developmental; they cannot support an unbiased confirmatory claim.
Gold-span substring matching	Paraphrased support may be missed; nDCG depends on representation granularity.
Selection among 12 candidates	Selection estimates are optimistic; paired tests on the same partition do not remove selection bias.
Different v1/v2 nominal maxima	Actual-token Pareto dominance is descriptive, not a controlled proof of v1 superiority.
No answer generation	Retrieval gains may not translate to answer token F1 or user-visible quality.
Current encoder/representation family	Negative routing evidence is scoped; it is not a universal impossibility result.

Paid work remains blocked

Gate	Status
Retrieval-only v2 artifacts, tests, reproduction	Complete
Answer prompt/model fingerprints and verified run-date rates	Pending
Explicit paid smoke authorization with call/cost ceiling	Not provided
Separate final held-out authorization	Not provided
Paid smoke / held-out answer evaluation	**NOT RUN**

[BLOCKED] No paid smoke test and no held-out answer-generation evaluation may run without explicit authorization. The current artifacts contain zero LLM calls.

Paper-safe conclusion

The tested RRF-based one-or-two-representation routing mechanism is not supported under the current encoder, representation family, datasets, and retrieval configuration. A fresh training-only recalibration selected adaptive paragraph retrieval and confirmed that retrieving beyond a shallow probe materially improves evidence recall. It also found a quality-efficiency trade-off against same-view fixed-k comparators. However, frozen v1 point-dominates the selected v2 operating point on mean recall and actual context, with no significant paired difference. The v2 replacement claim is therefore unsupported. Whether either method improves held-out answer quality remains unproven.

Run fingerprint: 7bbee5d1071c2e549be0804d391c4764265e29aae66fa2a695d7771100f362a7
V1 manifest SHA-256: 5c570ec2b4c52e1b0ea0f2152a6f8fb6f1fa1c1575e58cfb58bde2c434a54d05
V2 selected configuration SHA-256: c991560f562d12af41a176bad38ad9f89b03eb6b273360a67882cd59be0934d1